

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^+}^{\#1\alpha\beta\chi}$		
$\mathcal{K}_{0^+}^{\#1\dagger}$	$-\frac{\Upsilon_1 k^2}{2}$	0		
$\mathcal{K}_{0^+}^{\#1\dagger\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^+}^{\#1\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\alpha\beta}$
	$\mathcal{K}_{1^+}^{\#1\dagger\alpha\beta}$	$-\frac{\Upsilon_2 k^2}{2}$	$\frac{k^2}{2}\frac{{}^{(4)}\kappa_4}{\sqrt{2}}$	0
	$\mathcal{K}_{1^+}^{\#2\dagger\alpha\beta}$	$\frac{k^2}{2}\frac{{}^{(4)}\kappa_4}{\sqrt{2}}$	$\frac{\Upsilon_3 k^2}{2}$	0
	$\mathcal{K}_{1^+}^{\#1\dagger\alpha}$	0	0	0
	$\mathcal{K}_{1^+}^{\#2\dagger\alpha}$	0	0	0
			$\mathcal{K}_{2^+}^{\#1\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1\alpha\beta\chi}$
			$\mathcal{K}_{2^+}^{\#1\dagger\alpha\beta}$	$-\frac{\Upsilon_4 k^2}{2}$
			$\mathcal{K}_{2^+}^{\#1\dagger\alpha\beta\chi}$	0

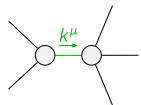
	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^+}^{\#1\alpha\beta\chi}$		
$\mathcal{J}_{0^+}^{\#1\dagger}$	$\frac{1}{\text{Det}(0^+)}$	0		
$\mathcal{J}_{0^+}^{\#1\dagger\alpha\beta\chi}$	0	0	$\mathcal{J}_{1^+}^{\#1\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2\alpha\beta}$
	$\mathcal{J}_{1^+}^{\#1\dagger\alpha\beta}$	$\frac{\Upsilon_3 k^2}{2\text{Det}(1^+)}$	$-\frac{k^2}{2}\frac{{}^{(4)}\kappa_4}{\sqrt{2}\text{Det}(1^+)}$	0
	$\mathcal{J}_{1^+}^{\#2\dagger\alpha\beta}$	$-\frac{k^2}{2}\frac{{}^{(4)}\kappa_4}{\sqrt{2}\text{Det}(1^+)}$	$-\frac{\Upsilon_2 k^2}{2\text{Det}(1^+)}$	0
	$\mathcal{J}_{1^+}^{\#1\dagger\alpha}$	0	0	0
	$\mathcal{J}_{1^+}^{\#2\dagger\alpha}$	0	0	0
			$\mathcal{J}_{2^+}^{\#1\alpha\beta}$	$\mathcal{J}_{2^+}^{\#1\alpha\beta\chi}$
			$\mathcal{J}_{2^+}^{\#1\dagger\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$
			$\mathcal{J}_{2^+}^{\#1\dagger\alpha\beta\chi}$	0

Abbreviations used in matrices

$$\begin{aligned} \Upsilon_1 &= 3\,{}^{(4)}\kappa_1 + {}^{(4)}\kappa_3 + {}^{(4)}\kappa_6 \,\&\& \Upsilon_2 = {}^{(4)}\kappa_3 - {}^{(4)}\kappa_6 \,\&\& \\ \Upsilon_3 &= {}^{(4)}\kappa_3 + {}^{(4)}\kappa_4 \,\&\& \Upsilon_4 = {}^{(4)}\kappa_3 + {}^{(4)}\kappa_6 \,\&\& \text{Det}(0^+) = -\frac{1}{2}k^2(3\,{}^{(4)}\kappa_1 + {}^{(4)}\kappa_3 + {}^{(4)}\kappa_6) \,\&\& \\ \text{Det}(1^+) &= -\frac{1}{8}k^4(2\,{}^{(4)}\kappa_3^2 + 2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^2 - 2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4){}^{(4)}\kappa_6) \,\&\& \text{Det}(2^+) = -\frac{1}{2}k^2({}^{(4)}\kappa_3 + {}^{(4)}\kappa_6) \end{aligned}$$

Lagrangian

$$\begin{aligned} &{}^{(4)}\kappa_1\partial_\beta\mathcal{K}^\delta_{\chi\delta}\partial^\chi\mathcal{K}^\alpha_{\alpha\beta} + {}^{(4)}\kappa_3\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta_{\alpha\chi} + {}^{(4)}\kappa_4\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta_{\beta\chi} + \\ &{}^{(4)}\kappa_6\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta_{\chi\alpha} + \frac{1}{2}{}^{(4)}\kappa_3\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta_{\beta\chi} + \frac{1}{2}{}^{(4)}\kappa_4\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta_{\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_0^{\#1\alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} ==$ $\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$	
$\mathcal{J}_1^{\#2\alpha} == 0$	3	$\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi} == 0$	
$\mathcal{J}_1^{\#1\alpha} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^\beta_{\beta\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}_{\beta} == \partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_2^{\#1\alpha\beta\chi} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} +$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^\delta_{\delta}{}^\epsilon + 3\,\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}_{\delta} ==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} +$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^\delta_{\delta}{}^\epsilon + 3\,\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}_{\delta}$	
Total # constraint(s):	12		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{4\,{}^{(4)}\kappa_3^2 + 4\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + 3\,{}^{(4)}\kappa_4^2 - 8\,{}^{(4)}\kappa_4\,{}^{(4)}\kappa_6 + 8\,{}^{(4)}\kappa_6^2}{(2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4({}^{(4)}\kappa_4 - 2\,{}^{(4)}\kappa_6) + 2\,{}^{(4)}\kappa_3({}^{(4)}\kappa_4 - {}^{(4)}\kappa_6))({}^{(4)}\kappa_3 + {}^{(4)}\kappa_6)}$

Resolved unitarity condition(s):

$$\begin{aligned} &({}^{(4)}\kappa_4 < 0 \,\&\& \\ &(({}^{(4)}\kappa_3 < -\frac{{}^{(4)}\kappa_4}{2} \,\&\& \frac{2\,{}^{(4)}\kappa_3^2 + 2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^2}{2\,{}^{(4)}\kappa_3 + 2\,{}^{(4)}\kappa_4} < {}^{(4)}\kappa_6 < -{}^{(4)}\kappa_3) \parallel (-\frac{{}^{(4)}\kappa_4}{2} < {}^{(4)}\kappa_3 < -{}^{(4)}\kappa_4 \,\&\& \frac{2\,{}^{(4)}\kappa_3^2 + 2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^2}{2\,{}^{(4)}\kappa_3 + 2\,{}^{(4)}\kappa_4} < {}^{(4)}\kappa_6 < -{}^{(4)}\kappa_3) \parallel \\ &({}^{(4)}\kappa_3 == -{}^{(4)}\kappa_4 \,\&\& {}^{(4)}\kappa_6 < -{}^{(4)}\kappa_3) \parallel ({}^{(4)}\kappa_3 > -{}^{(4)}\kappa_4 \,\&\& ({}^{(4)}\kappa_6 < -{}^{(4)}\kappa_3 \parallel {}^{(4)}\kappa_6 > \frac{2\,{}^{(4)}\kappa_3^2 + 2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^2}{2\,{}^{(4)}\kappa_3 + 2\,{}^{(4)}\kappa_4})))) \parallel \\ &({}^{(4)}\kappa_4 == 0 \,\&\& (({}^{(4)}\kappa_3 < 0 \,\&\& {}^{(4)}\kappa_3 < {}^{(4)}\kappa_6 < -{}^{(4)}\kappa_3) \parallel ({}^{(4)}\kappa_3 > 0 \,\&\& ({}^{(4)}\kappa_6 < -{}^{(4)}\kappa_3 \parallel {}^{(4)}\kappa_6 > {}^{(4)}\kappa_3)))) \parallel \\ &({}^{(4)}\kappa_4 > 0 \,\&\& (({}^{(4)}\kappa_3 < -{}^{(4)}\kappa_4 \,\&\& \frac{2\,{}^{(4)}\kappa_3^2 + 2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^2}{2\,{}^{(4)}\kappa_3 + 2\,{}^{(4)}\kappa_4} < {}^{(4)}\kappa_6 < -{}^{(4)}\kappa_3) \parallel \\ &({}^{(4)}\kappa_3 == -{}^{(4)}\kappa_4 \,\&\& {}^{(4)}\kappa_6 < -{}^{(4)}\kappa_3) \parallel (-{}^{(4)}\kappa_4 < {}^{(4)}\kappa_3 < -\frac{{}^{(4)}\kappa_4}{2} \,\&\& ({}^{(4)}\kappa_6 < -{}^{(4)}\kappa_3 \parallel {}^{(4)}\kappa_6 > \frac{2\,{}^{(4)}\kappa_3^2 + 2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^2}{2\,{}^{(4)}\kappa_3 + 2\,{}^{(4)}\kappa_4}))) \parallel \\ &({}^{(4)}\kappa_3 == -\frac{{}^{(4)}\kappa_4}{2} \,\&\& ({}^{(4)}\kappa_6 < \frac{2\,{}^{(4)}\kappa_3^2 + 2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^2}{2\,{}^{(4)}\kappa_3 + 2\,{}^{(4)}\kappa_4} \parallel {}^{(4)}\kappa_6 > \frac{2\,{}^{(4)}\kappa_3^2 + 2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^2}{2\,{}^{(4)}\kappa_3 + 2\,{}^{(4)}\kappa_4}))) \parallel \\ &({}^{(4)}\kappa_3 > -\frac{{}^{(4)}\kappa_4}{2} \,\&\& ({}^{(4)}\kappa_6 < -{}^{(4)}\kappa_3 \parallel {}^{(4)}\kappa_6 > \frac{2\,{}^{(4)}\kappa_3^2 + 2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4 + {}^{(4)}\kappa_4^2}{2\,{}^{(4)}\kappa_3 + 2\,{}^{(4)}\kappa_4})))) \end{aligned}$$